

RESEARCH ARTICLE

Is corporate greening beneficial? Exploring the relationship between perceived corporate environmental behavior and organizational citizenship behavior

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Abstract

Based on the social identity theory, this study investigates whether and how employee perceptions of corporate environmental behavior (CEB) influence the realization of an organization's internal value. Using employees' organizational citizenship behavior (OCB) as a direct agent for corporate internal value, the current study constructs a theoretical model of perceived CEB and OCB, with work meaningfulness and person–organization (P–O) value fit serving as mediator and moderator, respectively. After analyzing 235 questionnaire samples, we discovered that the perceived CEB positively influenced OCB via work meaningfulness and that this positive effect was strengthened by a greater P–O value fit. These findings suggest that in the process of motivating employees to implement OCB, firms should address the significance of internal driving roles of work meaningfulness and boundary-spanning roles of P–O value fit in order to enhance corporate internal value and correct managers' misconceptions about the value of CEB.

KEYWORDS

organizational citizenship behavior, perceived corporate environmental behavior, person–organization value fit, work meaningfulness

1 | INTRODUCTION

Corporate environmental behavior (CEB), also known as corporate green behavior, corporate environmental management practice, (Lin & Ho, 2010; Uhlaner et al., 2012), refers to a series of environmental measures that firms take proactively in response to external pressures or to reduce their negative impact on the environment (Algarni et al., 2022; Blok et al., 2015; Sarkar, 2008). It primarily consists of four aspects, including environmental strategy development, green production decision-making, green technology innovation, and environmental management implementation (Zhao et al., 2015). This is the

manifestation of corporate initiatives to fulfill social and environmental responsibilities (Li et al., 2017; Qin et al., 2019). Despite its contribution to both collective welfare and organizational effectiveness, CEB is typically unrewarded and optional for businesses (Boiral & Paillé, 2012; Ramus & Killmer, 2007). Thus, it is frequently characterized as “high cost-low benefit,” such as “complex, time-consuming, and unrecognized” (Boiral & Paillé, 2012) and “costly to self but beneficial for others” (Perc & Szolnoki, 2010), leaving decision-makers with the dilemma of “whether CEB must be truly implemented” (Perc & Szolnoki, 2010).

The dilemma compels researchers to examine whether CEB improves firms' internal value, which compresses both financial values, such as revenue and funding, and non-financial values, such as knowledge, market, and network-based benefits (De Silva et al., 2018). Some studies found that CEB improves firms' internal value, such as

List of Abbreviations: CEB, corporate environmental behavior; CSR, corporate social responsibility; OCB, organizational citizenship behavior; P–O, person–organization; R&D, research and development.

financial performance and competitiveness (Chuang & Huang, 2018; Miroshnychenko et al., 2017); however, other studies came to the opposite conclusion that CEB does not necessarily improve firms' financial performance and the majority of firms with CEB struggle to maintain profitability over time (Iraldo et al., 2009; Zeng et al., 2010). In business practice, many firms are hesitant to implement CEB because they perceive it to increase operating costs (Li et al., 2017; Orlitzky et al., 2011; Testa & D'Amato, 2017). Therefore, these firms may engage in pseudo-passive environmental behaviors in practice (Lyon & Maxwell, 2011; Perc & Szolnoki, 2010). Existing studies are inconclusive as to "whether" and "how" CEB achieves internal value and therefore cannot persuade and guide businesses to adopt proactive environmental behavior.

Moreover, the majority of prior studies have focused on financial performance as the internal value of firms (e.g., Chuang & Huang, 2018; Miroshnychenko et al., 2017), while ignoring the non-financial benefits, such as employees' positive attitudes and behaviors, which contribute to organizational effectiveness. With the absence of financial reward, employees' individual-level perceptions of corporate social responsibility (CSR) have positive effects on organizational identity (Shah et al., 2021), organizational trust (Serrano-Archimi et al., 2018), innovative behavior (Afridi et al., 2020), organizational citizenship behavior for the environment (Su & Swanson, 2019), and other employees' attitudes and behaviors. Once CEB is perceived by employees, it may spill over into employees' extra-role behaviors, thereby creating corporate value via micro-process mechanisms within the firm.

Different from outcome-oriented organizational-level variables such as financial performance, the current study focuses on employees' organizational citizenship behavior (OCB), a predictive and process-oriented individual-level variable (Chun et al., 2013; Kim & Gong, 2009). OCB is defined as "*individual behavior that is discretionary, not directly or explicitly recognized by the formal reward system, and that in the aggregate promotes the effective functioning of the organization*" (Organ, 1988, p. 4). OCB is a key predictor of short- and long-term economic performance despite its non-financial nature as an internal value from the perspective of employees (Chun et al., 2013; Kim & Gong, 2009; Podsakoff et al., 2014). No prior research has examined the relationship between employees' perceived CEB and OCB. In order to fill this gap, the current study links employees' perceived CEB with their OCB, which not only avoids the problem of the inconsistent relationship between perceived CEB and firm's internal value due to temporal or other factors but also addresses the "whether" and "how" questions.

The current study argues that when employees have a positive perception of CEB, it is important to extend this perception to their daily behaviors, including OCB. In conducting citizenship behavior, such as helping behavior, employees seek meanings in their work to strengthen their belief in making a difference in the lives of others (Supanti & Butcher, 2019). Work meaningfulness is defined as the worth of a work goal or purpose measured against an individual's ideals or standards (May et al., 2004; Spreitzer, 1995). While the implementation of CEB presupposes employee participation in engaging with and complying with various environmental requirements or

measures in the workplace, and so on (Boiral & Paillé, 2012), firms with CEB actually offer employees the chance to contribute to society (Glavas & Kelley, 2014). CEB serves as a social responsibility link between the individual, organization, and society (Aguinis & Glavas, 2019; Bailey et al., 2019) and is an ideal channel for individuals seeking meaningfulness (Brieger et al., 2020; Glavas & Kelley, 2014; Michaelson et al., 2014), which is distinct from organizational identity. For instance, Ashforth and Mael (1989) found that employees' identities are formed simply by being a part of the organization, even if there is no interaction between or within groups. Nevertheless, work meaningfulness is the specific psychological perception that ensures the internalization of identity into a more meaningful perception. In other words, organizational identity is a prerequisite for individuals to acquire a sense of meaningfulness; therefore, we believe that work meaningfulness is more likely to assist employees in transforming perceived CEB into daily work attitudes and behaviors. Steger et al. (2012) proposed that work meaningfulness could be used as a psychological variable to predict proactive work behaviors such as OCB. Therefore, we argue that the perceived CEB of employees extends to OCB via work meaningfulness.

Furthermore, according to social identity theory, inconsistencies or conflicts make it difficult for individuals to integrate the values, attitudes, and norms inherent to their various identities, and they may lead to hypocrisy or double standards (Ashforth & Mael, 1989). In line with the findings of Yang et al. (2020), we contend that the transition from perceived CEB to work meaningfulness is not invariant and may also be influenced by the alignment between a firm and its employees. In particular, they mentioned firm-employee alignment as a significant factor influencing employee hypocritical environmental behavior. Considering that the construction of perceived CEB to work meaningfulness is a flow from the organizational level to the individual psychological level, we introduce person-organization value fit (P-O value fit) as a boundary condition to investigate the crucial role of consistency in cementing their relationship. P-O value fit reflects the extent to which employees' value align with those of their organizational (Cable & DeRue, 2002; Hoffman & Woehr, 2006). The higher level of value fit employees perceive, the more likely they are to develop a stronger sense of identity and belonging to the organization (Edwards & Billsberry, 2010; Gould-Williams et al., 2015; Mi et al., 2020), thereby enhancing employees' understanding of CEB, which in turn influences the degree of work meaningfulness they perceive (Edwards & Billsberry, 2010). In order to better explain the relationship between employees' perceived CEB and work meaningfulness, the current study introduces P-O value fit as a moderator.

In summary, our study constructs a moderated mediation model, with work meaningfulness serving as a mediator between perceived CEB and OCB and P-O value fit serving as a moderator between perceived CEB and work meaningfulness. Reviewing the relevant literature and developing hypotheses, the subsequent section of the current study describes the methodology we employed and the results. Finally, we discuss these findings in terms of their theoretical contributions and practical implications, as well as their research limitations and future directions.

2 | LITERATURE REVIEW AND HYPOTHESIS DEVELOPMENT

2.1 | Perceived CEB and employee OCB

Environmental strategy development, environmental management, green technology innovation, and green production decision-making are typically the primary means by which businesses implement environmental behavior (Zhao et al., 2015). Among these, environmental strategy formulation is the long-term and comprehensive planning of a firm's green and sustainable development goals, such as promoting the development of renewable energy (which falls within the scope of business operations) (Xie et al., 2020). Environmental management refers to the behaviors directly related to environmental protection, such as establishing environmental management systems and strengthening environmental education and training (Zhao et al., 2015). Additionally, green technological innovation refers to technological innovations that aim to protect the environment, such as terminal emission reduction innovation and energy-saving equipment innovation (Liao & Tsai, 2019). Lastly, green production decision refers to production decisions in the production process aimed at energy conservation, consumption reduction, and pollution reduction, such as switching to cleaner inputs (Sharma, 2000; Xepapadeas, 1992). Firms encourage employees to fully participate in their environmental behaviors by setting green goals, developing environmental systems, implementing technological innovation, and making production decisions, which not only helps firms to form good reputation and social status (Hoffman, 2001) but also demonstrates to other stakeholders, including employees, the firms' positive value orientation and correct ethical awareness (Parsa et al., 2015; Venhoeven et al., 2013).

According to social identity theory, identity is a perceptual-cognitive construct (Ashforth & Mael, 1989), which is simply achieved by employees' affiliation with an organization, and that behaviors and emotions are merely potential antecedents or outcomes of identification (Foote, 1951; Gould, 1975). However, given that "*identification is defined as organization-specific, internalization and commitment may not be*" (Ashforth & Mael, 1989, p. 23), we propose that the perception of a deeper meaning, or work meaningfulness, is the only driver of further identification. In contrast to the corporate-level conceptualization of CEB, the current study adopts the CEB from the employee perception perspective, which is an individual-level conceptualization (Glavas & Kelley, 2014). On the one hand, individual-level perception contributes to the investigation of how employees can enhance identity transfer between the organizational and individual psychological levels. For instance, employees' perceptions of the firm's ethical values strengthen their identification with the firm (Liu & Lin, 2016). On the other hand, by perceiving CEB, employees gain a deeper understanding of "identity" and can choose whether and how to translate the information provided by CEB to the psychological and behavioral level of the individual. Perceptually, it is therefore easier to comprehend how CEB influences individual-level variables such as employee OCB. In other words, the perceived effect of CEB on

employee attitudes must demonstrate the value of CEB (Glavas & Kelley, 2014). Therefore, we define perceived CEB as the extent to which employees perceive environmental strategy development, environmental management, green technology innovation, and green production decision-making to be implemented.

As mention previously, OCB is individual discretionary behavior that is not directly or explicitly linked to the organization's formal reward system (Organ, 1988) and is characterized by extra-role behavioral characteristics such as altruism, ethics, and responsibility (Chen et al., 2005; Podsakoff et al., 2014). In the process of developing and implementing environmental strategies, firms deliver ethics, altruistic, and sustainable values to their employees in a variety of ways (Lülfes & Hahn, 2013; Zhao et al., 2015), which may promote employees' OCB (De Roeck & Farooq, 2018; Khan & Khan, 2022). In the meantime, the implementation of environmental management systems, green technological innovation, and green production decision-making frequently requires the participation and advancement of employees from various departments such as production, research and development (R&D), and finance (Boiral & Paillé, 2012; Ramus, 2001). For instance, businesses must encourage R&D or knowledge-based employees to actively work on technological innovation of end-of-pipe emission reduction or energy-saving equipment, which requires the cooperation of functional departments such as finance and human resources. Then, by participating in CEB, employees gain a deeper understanding of the norms and requirements such as environmental responsibility, ethics and altruism that are emphasized by firms (Hansen et al., 2011), thereby enhancing their knowledge and comprehension of CEB (Afsar et al., 2018; Weick, 1995). As a result, employees are more empathetic, concerned with organizational development, and willing to participate in processes that contribute to organizational shaping and further promote their OCBs (Cullen et al., 2003). Therefore, we propose the following:

Hypothesis 1. Employees' perceived CEB has a positive effect on their OCBs.

2.2 | The mediating role of work meaningfulness

Work is the foundation for self-realization (Jang, 2001), and meaning-seeking satisfies the internal motivation of individuals in the workplace (Chalofsky & Krishna, 2009; Jung & Yoon, 2016). Work meaningfulness reflects a connection or interrelationship between an individual's inner world and the workplace's external manifestations (Cartwright & Holmes, 2006). It is also viewed as a compatibility between job role requirements and individual beliefs, values, and behaviors (Brief & Nord, 1990; Spreitzer, 1995). We speculate that work meaningfulness reflects a well-defined link between an individual's internal beliefs and external role expectations. It is impacted by CSR (Aguinis & Glavas, 2019), such as the deployment of CEB, and develops a greater drive to generate positive behaviors, such as OCB (Supanti & Butcher, 2019). The justifications for using this approach are summed up in three points.

First, we propose that the economic and non-economic contributions made by firms with CEB to external stakeholders such as shareholders, customers, and communities (Wang et al., 2018) strengthen the firms' social reputation (Hoffman, 2001), resulting in shared collective achievement among employees. According to social identity theory, shared collective performance enables employees to associate with firms because of shared identities and beliefs when perceiving environmental behaviors of the firms (Hogg & Terry, 2000). In comparison to out-groups, employees find that CEB provides salient moral attributes that distinguish them from other firms (Kim et al., 2010). Then they regard the group to which they belong as valuable and unique in accordance with mainstream social values (Tajfel & Turner, 1979), thereby reinforcing their in-group membership and status. Consequently, employees continue to strengthen their sense of belonging, job engagement and organizational commitment to the firms, as well as their work meaningfulness (Dik et al., 2013; Jung & Yoon, 2016; Renn & Vandenberg, 1995).

Second, in addition to fostering a sense of belonging, perceived CEB may also assist employees in activating self-esteem mechanisms to boost work meaningfulness. According to social identity theory, individuals who regard themselves as valuable in-group members at work are more likely to have higher self-esteem (Ashforth & Mael, 1989), which contributes to a sense of meaningfulness at work. Studies have shown that collective experiences and accomplishments impact individual self-esteem (Crocker & Park, 2004; Leary & Downs, 1995). Therefore, we suggest that when employees see firms' readiness to implement environmental behaviors for the benefit of society as a whole, the collective success they generate continues to validate their internal value and contribution as in-group members engaged in CEB (Ashforth & Mael, 1989). This protection and reinforcement of a positive self-perception enables individuals to sustain greater levels of self-esteem (Crocker & Park, 2004; Gecas, 1991; Leary & Downs, 1995; Rosso et al., 2010), which in turn inspires work meaningfulness.

Third, self-transcendence refers to linking the ego to an entity that is bigger than the self or the world (Maslow, 1971). Wide-ranging CEB initiatives not only contribute to the firm's long-term competitive advantage but also effectively improve the environmental quality and safeguard the safety and health of the surrounding community's citizens. By engaging in the specific work of CEB, employees genuinely feel a connection between what they are doing and the larger aims that transcend their own lives (Rosso et al., 2010). On the one hand, employees believe that work affords them opportunity to positively impact society or the world, allowing them to experience higher spiritual values and the meaning of their work (Lips-Wiersma, 2002). Alternatively, when employees' self-interest conflicts with organizational environmental requirements, they choose to comply with corporate environmental goals or job role requirements to ease their anxiety produced by the conflicts, which creates a greater degree of control over employees (Kreiner et al., 2006). Consequently, employees seek self-transcendence in a feeling of self-abnegation (Rosso et al., 2010), thereby experiencing a greater sense of work meaningfulness.

Work meaningfulness has been demonstrated by previous research to offer employees spiritual nutrition and work enjoyment

(Bowie, 1998; Leunissen et al., 2018) and to predict positive attitudes and behaviors (Vogel et al., 2020), including OCBs (Chen et al., 2011; Maharaj & Schlechter, 2007). Employees tend to put more effort into work, commit to organizational goals and values (Jung & Yoon, 2016), and gain more recognition through extra-role work (Ko et al., 2018). Therefore, the acquisition of work meaningfulness is conducive to building a bridge between perceived CEB and employee's OCB. Therefore, we propose the following:

Hypothesis 2. Work meaningfulness mediates the positive relationship between perceived CEB and employees' OCB.

2.3 | The moderating role of person–organization value fit

According to social identity theory, identity emerges even in the absence of effective leadership and member interaction, although but it does not necessarily grow or become more internalized continuously. One important factor that may increase identity is the similarity between an individual and his or her organization, which is a crucial element in establishing identity (Ashforth & Mael, 1989). In general, if an employee perceives a greater degree of value fit between himself/herself and his/her organization, the match between perceived job role requirements and individual values will be greater, and, as a result, in turn, employees' work becomes more meaningful. Therefore, we use P-O value fit as the moderator between employees' perceived CEB and work meaningfulness.

Specifically, P-O value fit allows employees to be socially categorized based on their similarity to the organization (Tajfel, 1982). Employees with a high P-O value fit degree share cognitive and behavioral traits with the firm. Thus, people are able to comprehend the objective of organizational management (Cable & DeRue, 2002; Hoffman & Woehr, 2006), and their sense of belonging to the firm grows. Then, the more employees perceive the firm's favorable intent toward environmental protection in the process of implementing CEB, the more organizational expectations and their personal psychological needs are satisfied, and the greater the likelihood that they will sense work meaningfulness (Edwards & Cable, 2009). Furthermore, P-O value fit assists employees in identifying and validating their beliefs, allowing them to receive clear job expectations, hence enhancing the effect of CEB on work meaningfulness. A high P-O value fit match helps employees to completely confirm that "CEB is consistent with employees' values or beliefs," thereby satisfying employees' spiritual value needs (Afsar & Badir, 2016). Consequently, with explicit role expectations, employees are more likely to contribute to the attainment of corporate value goals (Albrecht, 2015) and realize their own work meaningfulness through these contributing behaviors. In conclusion, we propose, on the basis of social identity theory, that P-O value fit is as an important aspect in strengthening the identity formation process. We argue that P-O value fit supports the spiritual shift from employees' perceived CEB



to work meaningfulness through social categorization and belief validation.

Conversely, when the P-O value fit is low, employees may be subject to contradictory value standards (Ashforth & Mael, 1989). Such contradictory value standards induce psychological dissonance in employees and make it harder for them to feel moral support from the firms, thus diminishing the psychological connotation that employees establish a united front with the firm through their involvement in CEB. A low P-O value fit, on the other hand, indicates that employees perceive their own values or organizational values to be difficult to support (Cable & DeRue, 2002; Hoffman & Woehr, 2006), resulting in less positive work-related psychological experiences (self-esteem, belongingness, and self-transcendence) for employees as “organizational people,” which ultimately results in less work meaningfulness. Therefore, we hypothesize the following:

Hypothesis 3. The relationship between employees' perceived CEB and work meaningfulness is positively moderated by higher levels of P-O value fit, and negatively moderated by lower levels of P-O value fit.

As stated previously, Hypothesis 3 reveals that P-O value fit moderates the relationship between perceived CEB and work meaningfulness. Combining Hypotheses 3 and 2, the P-O value fit plays a moderating role in the mediating effect of work meaningfulness. In particular, based on social identity theory, the higher the P-O value fit is, the greater the likelihood that employees perceive organizational support for their own values and beliefs (Afsar & Badir, 2016). They strengthen their identification with the organization and have a stronger knowledge of their job expectations, which helps the spiritual transformation between perceived CEB and work meaningfulness, and ultimately experiences greater work meaningfulness. Moreover, employees who derive pleasure and spiritual fulfillment from their work are internally driven and more inclined to initiate OCB. On the contrary, when the P-O value fit is lower, it is difficult for employees to experience spiritual support from the organization, which leads to a lower sense of identity as in-group members, thereby diminishing the experience of work meaningfulness and reducing the internal value transformation from perceived CEB to employees' OCB. Thus, Hypothesis 4 is proposed and a model for hypotheses is constructed, as depicted in Figure 1.

Hypothesis 4. P-O value fit moderates the relationship between perceived CEB and OCB through work meaningfulness, such that the indirect effect is stronger with higher P-O value fit than with lower P-O value fit.

3 | METHODOLOGY

3.1 | Sample and data collection

In this study, we conducted a paper-based questionnaire survey in three Chinese manufacturing firms from Jiangsu, Henan and Shandong provinces, respectively. We contacted the human resources managers of these firms, and presented the research goal and method of data collection. With a comprehensive understanding of the research, the managers assisted in recruiting volunteers for their respective firms. We recruited 300 individuals for the distribution of questionnaires and collecting of data. Notably, to ensure the validity and reliability of the data collected, we provided respondents with thorough instructions on how to complete the questionnaires prior to the survey, emphasizing that the information would be kept totally confidential and used exclusively for research purposes.

The survey was conducted in two phases, with a 2-month interval, to prevent the possibility of common method bias resulting from data collecting at a single time point (Podsakoff et al., 2003). At the phase one survey, a questionnaire containing personal information, perceived CEB and P-O value fit questions was distributed to 300 participating employees, of whom 264 responded. Two months later, for the phase two survey, we issued a second questionnaire to employees who had completed the phase one survey, asking them to evaluate the work meaningfulness and OCB of their employment.

We matched the questionnaire respondents of the two phases based on the job numbers provided by the employees. After removing incomplete questionnaires, 235 valid samples are identified, revealing a return rate of 78%. The descriptive statistical analysis of the sample indicates that 68.9% of respondents were male, 48.1% were married, and 49.8% were between the ages of 26 and 35. Moreover, 55.7% of respondents had been with their present firm for less than 3 years, 55.7% held a bachelor's degree or above.

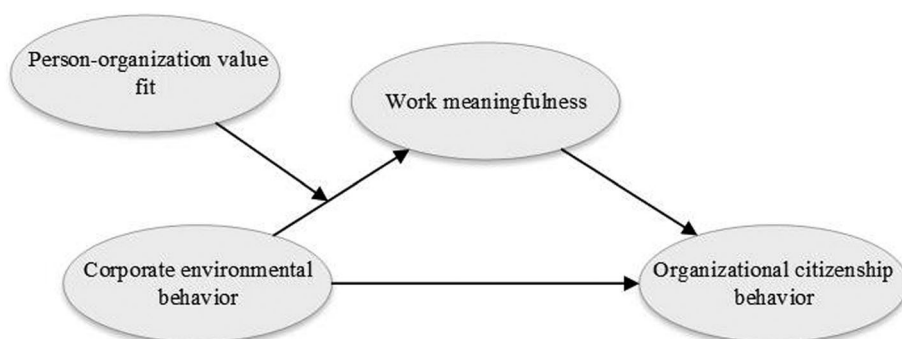


FIGURE 1 The theoretical model

3.2 | Measures

The authoritative and mature scales were employed for measuring in this study. To ensure that the questionnaire design and statements were adequate, we translated them into Chinese following the method suggested by Brislin (1986). In addition, we modified the wording of the questionnaires to verify that the respondents understood them correctly and that the content was accurate. All items were measured by a 5-point Likert scale, ranging from 5 “*strongly agree*” to 1 “*strongly disagree*.”

3.2.1 | Perceived corporate environmental behavior

We adapted a 17-item scale developed by Zhao et al. (2015) to measure perceived CEB. Sample items include “transferring to clean industry” and “buying clean materials.” The Cronbach's alpha of the scale was .95.

3.2.2 | Work meaningfulness

We selected a 3-item scale developed by Spreitzer (1995) to measure work meaningfulness. Sample items include “the work I do is very important to me” and “the work I do is meaningful to me.” The Cronbach's alpha of the scale was .89.

3.2.3 | Organizational citizenship behavior

We applied a 15-item scale simplified from Hui et al. (1999) by Farh et al. (1997) to measure OCB. Sample items include “I am willing to help new colleagues to adapt to the work environment,” “I am eager to tell outsiders good news about the firm and clarify their misunderstandings,” and “I take one's job seriously and rarely make mistakes.” The Cronbach's alpha of the scale was .87.

3.2.4 | Person–organization value fit

A 3-item scale adapted from Cable and DeRue (2002) by Hoffman et al. (2011) was used to assess P-O value fit. Sample items include

“my personal values match my organizational values and ideals” and “the things that I value in life are similar to my organizational values”. The Cronbach's alpha of the scale was .77.

3.2.5 | Control variables

Finally, we controlled for the effects of gender, marriage, age, tenure, and education, since these variables are related to perceived CEB, work meaningfulness, OCB, and/or P-O value fit (Indartono & Wulandari, 2014; Lee, 2006; Özen & Küskü, 2009; Situ & Tilt, 2018).

4 | RESULTS

4.1 | Validity analysis

The Cronbach's alpha coefficients for perceived CEB, work meaningfulness, OCB and P-O value fit were all greater than .7, indicating that the study's scale had acceptable reliability. Next, we ran a series of confirmatory factor analyses (CFAs) to provide empirical evidence of the variables' distinctiveness. We compared the fit indexes of the baseline four-factor model with the five alternative models by using Mplus 8.0 (Anderson & Gerbing, 1988). As shown in Table 1, the four-factor model fit indexes well ($\chi^2/df = 1.64$, CFI = 0.91, TLI = 0.90, RMSEA = 0.05), and significantly better than alternative models. This result indicates that not only did our model have strong convergent validity but also good discriminant validity; hence, the hypothesis is tested.

4.2 | Common method variance issue

The common method variance (CMV) is a potential threat to the validity of research findings (Podsakoff et al., 2003). Comparing (1) the measurement model with the addition of the unmeasured latent CMV factor ($\chi^2/df = 1.59$, CFI = 0.92, TLI = 0.91, RMSEA = 0.05) and (2) the same measurement model without the CMV factor ($\chi^2/df = 1.64$, CFI = 0.91, TLI = 0.90, RMSEA = 0.05), we found that the changes of the fit indexes were not significant ($\Delta CFI = 0.01$, $\Delta TLI = 0.01$, $\Delta RMSEA = 0.00$, respectively), indicating that the model

TABLE 1 Measurement model and alternative model comparison

Model	Factors details	Chi ²	df	Chi ² /df	CFI	TLI	RMSEA
Model 1	Four factors: CEB, WM, OCB, POF	1066.54	650	1.64	0.91	0.90	0.05
Model 2	Three factors: CEB, WM + POF, OCB	1246.16	653	1.91	0.87	0.85	0.06
Model 3	Three factors: CEB + OCB, WM, POF	1315.84	653	2.02	0.85	0.84	0.07
Model 4	Two factors: CEB, WM + OCB + POF	1267.52	654	1.94	0.86	0.85	0.06
Model 5	Two factors: CEB + WM + POF, OCB	1247.80	654	1.91	0.86	0.85	0.06
Model 6	One factor: CEB + WM + OCB + POF	1480.63	655	2.26	0.81	0.80	0.07

Abbreviations: OCB, organizational citizenship behavior; PCEB, perceived corporate environmental behavior; POF, person–organization value fit; WM, work meaningfulness.

was not enhanced by the addition of the common method factor and suggesting that there was no significant common method bias problem (Cheung & Rensvold, 2002).

4.3 | Descriptive statistics

Table 2 shows the descriptive statistics, correlation coefficients, and reliability coefficients for each variable. In addition, perceived CEB was significantly positively correlated with OCB ($r = 0.51$, $p < .001$), and work meaningfulness and P-O value fit ($r = 0.27$,

$p < .001$) had a significant positive correlation, tentatively validating the hypothesis.

4.4 | Hypothesis testing

4.4.1 | Mediation

Model coefficients for the conditional process analysis are presented in Table 3. The results show that perceived CEB was positively associated with OCB ($\beta = .41$, $p < .001$), thus supporting Hypothesis 1.

TABLE 2 Descriptive statistics and correlation analysis

Variables	Mean	SD	1	2	3	4	5	6	7	8	9
Gender	0.31	0.46									
Marriage	0.52	0.50	0.09								
Age	2.00	0.90	−0.18**	−0.62***							
Tenure	1.78	1.15	−0.04	−0.54***	0.77***						
Education	3.47	0.99	0.06	0.17*	−0.21**	−0.19**					
PCEB	3.98	0.67	0.10	−0.09	0.10	0.16*	0.05	(0.95)			
OCB	3.49	0.53	0.06	−0.04	0.09	0.11	−0.07	0.51***	(0.87)		
WM	3.87	0.79	0.14*	0.05	0.13*	0.18**	0.01	0.33***	0.46***	(0.89)	
POF	3.94	0.78	0.08	0.01	−0.01	0.07	−0.02	0.24***	0.43***	0.27***	(0.77)

Notes: $N = 249$. Reliability coefficients are reported along the diagonal. Gender: male (0) and female (1); marriage: married (0) and unmarried (1); age: Under 18 years old (1); 19–25 years old (2); 26–35 years old (3); 36–44 years old (4); 45 years old and above (5); tenure: 0–3 years(1), 3–5 years(2), 6–10 years(3), 11–15 years(4), 16–20 years (5), >20 years (6); education: Junior high school and below (1), High school and secondary school (2), Junior college (3), Undergraduate (4), Master (5), Doctor and above (6).

Abbreviations: OCB, organizational citizenship behavior; PCEB, perceived corporate environmental behavior; POF, person–organization value fit; WM, work meaningfulness.

* $p < .05$. ** $p < .01$. *** $p < .001$.

TABLE 3 Model coefficients for the conditional process analysis

Model	Organizational citizenship behavior			Work meaningfulness		
	Model 1	Model 2	Model 3	Model 5	Model 6	Model 7
Predictor	B(S.E.)	B(S.E.)	B(S.E.)	B(S.E.)	B(S.E.)	B(S.E.)
Constant	3.57(0.17) ***	3.7(0.18) ***	3.74(0.16) ***	3.12(0.27) ***	3.09(0.27) ***	3.09(0.26) ***
Gender	0.02(0.07)	0.01(0.07)	−0.02(0.06)	0.2(0.11)	0.18(0.1)	0.19(0.1)
Marriage	0.06(0.08)	−0.08(0.08)	−0.03(0.07)	0.4(0.12) **	0.39(0.12) **	0.39(0.12) **
Age	0.03(0.06)	−0.01(0.06)	0(0.05)	0.13(0.09)	0.15(0.09)	0.15(0.09)
Tenure	0(0.04)	−0.01(0.04)	−0.02(0.04)	0.11(0.07)	0.09(0.06)	0.09(0.06)
Education	−0.05(0.03)	−0.04(0.03)	−0.05(0.03)	0(0.05)	0.01(0.05)	0.01(0.05)
PCEB	0.41(0.05) ***		0.23(0.04) ***	0.36(0.07) ***	0.31(0.07) ***	0.31(0.07) ***
WM		0.32(0.04) ***	0.33(0.05) ***			
POF					0.2(0.06) **	0.20(0.06) **
PCEB × POF						0.21(0.08) **
ΔR^2			0.15***	0.09***	0.03**	0.02**
R^2	0.28	0.22	0.37	0.18	0.21	0.24
F	14.53	10.68	18.92	8.29	8.83	8.85

Abbreviations: OCB, organizational citizenship behavior; PCEB, perceived corporate environmental behavior; POF: person–organization value fit; S.E., standard error; WM, work meaningfulness.

* $p < .05$. ** $p < .01$. *** $p < .001$.

Moreover, the results of the analysis revealed that perceived CEB was positively associated with work meaningfulness ($\beta = .36$, $p < .001$) and that work meaningfulness was positively associated with OCB ($\beta = .32$, $p < .001$). Finally, we examined the total, direct and indirect effect of perceived CEB on OCB. Table 4 demonstrates that the indirect effect was significant (indirect effect = 0.08, 95% CI = [0.04, 0.13]), and there was an indirect relationship. These results support Hypothesis 2 that work meaningfulness partially mediated the effect of perceived CEB on OCB.

4.4.2 | Moderation

As shown in Table 3, the non-standard regression coefficient of the interaction term between perceived CEB and P-O value fit was 0.21 ($p < .01$), $\Delta R^2 = 0.02$ ($p < .01$), and the R -squared was highly variable, indicating that the P-O value fit positively and significantly moderates the relationship between perceived CEB and work meaningfulness. In addition, as shown in Figure 2, the positive relationship between work meaningfulness and perceived CEB was higher when P-O value fit was high ($\beta = .47$, $p < .001$) than when the P-O value fit was low ($\beta = .14$, $p > .01$). Hence, Hypothesis 3 was supported.

TABLE 4 The mediation model showing total, direct, and indirect effect of perceived CEB on OCB ($n = 249$)

	Effect	S.E.	LLCI	ULCI
Direct effect				
PCEB→OCB	0.331	0.045	0.242	0.420
Indirect effect				
PCEB→WM→OCB	0.081	0.024	0.039	0.133
Total effect				
PCEB→OCB	0.412	0.046	0.322	0.502

Abbreviations. OCB, organizational citizenship behavior; PCEB, perceived corporate environmental behavior; POF, person-organization value fit; S.E., standard error; WM, work meaningfulness.

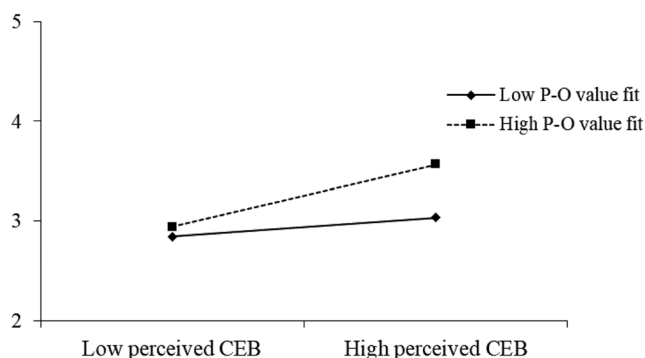


FIGURE 2 P-O value fit moderates the effect of perceived corporate environmental behavior (CEB) on work meaningfulness

TABLE 5 Results of moderated mediation analysis

Moderator values	Effect	S.E.	LLCI	ULCI
Low POF	0.032	0.020	−0.011	0.070
High POF	0.106***	0.038	0.038	0.185
Difference	0.074**	0.040	0.004	0.159

Abbreviations. OCB, organizational citizenship behavior; PCEB, perceived corporate environmental behavior; POF, person-organization value fit; S.E., standard error; WM, work meaningfulness.

4.4.3 | Moderated mediation

The results of verifying the mediated moderating effect in Hypothesis 4 are shown in Table 5. The index of moderated mediation was 0.047 (95% CI [0.003, 0.102]), indicating that there was a mediating effect of the model being moderated. The mediating effect was not significant when the P-O value fit was low (Effect = 0.032, 95% CI [−0.011, 0.070]), while it was significant when the P-O value fit was high (Effect = 0.106, 95% CI [0.038, 0.185]). At the same time, the difference between the two groups was significant (Effect = 0.073, 95% CI [0.004, 0.159]). In summary, P-O value fit moderated the mediating role of work meaningfulness in the relationship between perceived CEB and OCB. Therefore, Hypothesis 4 was supported.

5 | DISCUSSION

Is corporate greening beneficial to the internal value of firms? In the current study, we investigate the decision dilemma of “whether it is necessary for firms to implement CEB environmental behavior and how this impacts the internal value of the firms” and, from the employees’ perspective, selects employee OCB, a processual, individual-level indicator variable that predicts both short- and long-term economic performance, to measure the corporate internal value. What’s more, this study explores the mechanism of perceived CEB on employee OCB from the perspective of employee perception. We introduced social identity theory and constructed a moderated mediating effect model to examine the direct relationship between employees’ perceived CEB and their OCB. This explains that work meaningfulness is an important way to determine the significance of work meaningfulness in transforming CEB into internal value as a mediating variable. Moreover, we examined the boundary role of P-O value fit to answer the questions of “whether” and “why” CEB attains internal value.

Implementing environmental behaviors of firms motivates employees to actively engage in OCB. This finding is consistent with prior research indicating that CEB components, such as corporate environmental policies and environmental management practices, have a positive impact on OCB (Paillé et al., 2013; Raineri & Paillé, 2016). In addition, the current study indicated that work meaningfulness is a means by which employees realize corporate values. They internalize perceived CEB into spiritual meanings and values such as belonging, self-esteem, and self-transcendence, thereby

translating into employee OCB, which ultimately leads to the realization of their firms' internal values. Work meaningfulness, as the fulfillment of employees' high-level spiritual needs, derives from CEB and influences OCB. CEB promotes employees' work meaningfulness as a strategy of achieving CSR. Consistent with earlier research, perceived CSR boosts the meaningfulness of employees' work (Supanti & Butcher, 2019). Also, employees with a greater sense of work meaningfulness are more likely to go above and beyond their duty requirements, or to improve their OCB. Previous research has also demonstrated that employees' work meaningfulness has a positive impact on their OCB (Chen et al., 2011; Maharaj & Schlechter, 2007; Malik et al., 2011). This study concludes by demonstrating that P-O value fit is a boundary condition for strengthening the mediating effect of work meaningfulness. More so than employees with a lower P-O value fit, employees with a higher P-O value fit reinforce the facilitative effect of work meaningfulness on perceived CEB and OCB. This finding indicates that P-O value fit enables employees to comprehend the impact of CEB on their attitudes and behaviors.

The current research contributes to the existing literature by adapting social identity theory to examine the transformation of organizational-level to individual-level behaviors. The theoretical findings also have implications for practitioners in the implementation of greening decisions at the corporate level.

5.1 | Theoretical implications

The current study contributes to the field of organizational behavior, CSR, and employees' OCB in three primary aspects.

Firstly, from the standpoint of social identity theory, we investigate the probability of perceived CEB spillover to employees' daily work performance (e.g., OCB). The majority of prior research has focused on the impact of perceived CEB on environmental performance (Cheema et al., 2020; Robertson & Barling, 2017; Yuriev et al., 2018), but not on the effect of CEB on the corporate internal value. To address this gap in literature, we investigate how perceived CEB influences employee OCB in order to explicate the impact mechanism of CEB on internal value. Incorporating CEB and OCB, the current study not only contributes to the literature on CEB and employee OCB, but also addresses the questions of "whether" and "how" CEB influences internal value. On the one hand, our study overcomes the decentralized limitation of prior CEB research that only focused on specific dimensions (e.g., corporate environmental protection policies and environmental management practices) (Paillé et al., 2013; Raineri & Paillé, 2016) and integrates environmental strategy development, environmental management, green technology innovation, and green production decisions into a unified research framework. It expands the application of CEB in organizational behavior research and demonstrates the viability of transforming CEB into firm internal value. In contrast to prior research on measuring firm value with outcome-oriented organizational-level variables, such as financial performance (Chuang & Huang, 2018; Miroshnychenko et al., 2017; Zeng et al., 2010), the current study demonstrates the

rationale for viewing employee OCB as a process-oriented individual-level variable that predicts firm internal value and examines the psychological path of CEB in promoting employee OCB. In light of these findings, our study enriches the literature on the "trickle-down" effect of behavioral transfer from positive behavior at the organizational level to extra-role behavior at the individual level and provides theoretical support for convincing firms to adopt proactive environmental behavior.

Second, our study aims to explore, from a spiritual supply-demand viewpoint, how CEB converts into corporate internal value. By introducing the mediating variable of work meaningfulness, we uncovered a major psychological mechanism of perceived CEB spillover to employee OCB, thereby filling the gap left by earlier research on spiritual needs in the field of CEB research. Considering that CEB must be perceived by employees in order to influence their attitudes (Glavas & Kelley, 2014), the variable of perceived CEB was chosen. Consequently, based on social identity theory, we selected the spiritual variable, namely, work meaningfulness, as the "spiritual bridge" between employees' perceived CEB and OCB in order to establish the spiritual connection between firm-work-employee. By incorporating perceived CEB, work meaningfulness, and employee OCB into a unified research framework, this study dynamically presents the micro-processes that occur within firms to account for the CEB-corporate performance relationship, complementing and explaining the psychological mechanisms of CEB enhancing corporate internal value. In the meantime, our study provides support for the psychological hypotheses of social classification, social comparison, and positive distinction in social identity theory, thereby enriching the research field of social identity theory and providing theoretical support for increasing the supply of spiritual resources for firms and enhancing the psychological demands placed on employees.

Lastly, the current study highlighted boundary conditions that support the transformation from perceived CEB to work meaningfulness. We suggest that P-O value fit is a potent means of preventing double standards among employees and so steadily promoting the transformation of perceived CEB to work meaningfulness. This procedure provides a detailed description of how and when CEB impacts the work meaningfulness of employees. We emphasize the boundary role of P-O value fit from the perspective of the interaction between a firm and its employees' values, and the research conclusions explain "when" by addressing the questions of "whether" and "how."

5.2 | Practical implications

This research provides firms and decision-makers with practical implications for effectively implementing environmental behavior and managing the relationship between firms and their employees.

First, it is necessary for firms to strengthen training on environmental awareness for decision-makers so that they recognize CEB is not a zero-sum game, but rather an effective tool for creating corporate internal value. Firms should actively identify and promote managers who are aligned with the direction of sustainable development

of organizations and ultimately urge them to formulate appropriate policies and practices.

Second, firms should hire individuals who align with their organizational values and invest in social responsibility and ethical training for their staff. Firms educate their employees on the value of their social responsibilities, such as environmental protection, so as to enhance their organizational identity and foster their work meaningfulness. In addition, firms could increase employees' sense of belonging through group activities, hence increasing their OCB adoption propensity.

Third, corporate decision-makers should be aware that the implementation of CEB improves the firm's financial performance by improving employee OCB. In other words, corporate decision-makers are encouraged to set an example, vigorously promote and publicize the value and significance of CEB, and assist employees in understanding the significance of their work through initiatives such as developing environmental protection strategies, implementing green production decisions, initiating green technological innovation, and improving environmental management. These initiatives encourage employees to implement OCB, hence creating internal value for the firm.

Lastly, in the process of implementing environmental behaviors, corporate decision-makers should focus on developing an effective internal environmental management system, such as minimizing pollution emissions, enhancing environmental protection education, and training staff. Simultaneously, they must increase the innovation of corporate production processes and equipment and promote green decisions, establish the firm's green environmental reputation, and meet the environmental expectations of employees in full, which will have a continuous positive effect on the mental strength of employees. This series of measures enables employees to reevaluate their organizational membership and actively participate in the future building of the organization, thus maximizing employee participation in adopting OCB and increasing value for the firm.

5.3 | Limitations and future research directions

This research has the following limitations. Firstly, our study reveals the relationship between perceived CEB and employees' OCB with the mediating variable of work meaningfulness, yet lacks sufficient investigation of other mediators. Future research may examine additional mediating variables, such as emotional variables, to understand their roles in the relationship between perceived corporate behavior and employee behavior and to provide insights on the differential effects of these mediators.

Second, our study primarily evaluates CEB from four aspects (environmental strategy development, environmental management, green technology innovation, and green production decisions) without undertaking an empirical examination of the dimensions' interaction. Therefore, future research may explore the interaction between the dimensions of CEB and OCB, which would contribute to a deeper comprehension of the significance of CEB to firms.

Finally, our study collected data from three Chinese firms. Due to the intersection effects of complex background factors, such as the economic position, degree of pollution, and cultural background of the regions, the limited sample size may introduce bias into the data distribution. Larger data sets with more diverse populations could be employed in the future to further validate the findings of the current study.

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CONFLICT OF INTEREST

We have no conflicts of interest.

ETHICS STATEMENT

We comply with ethical standard.

AUTHOR CONTRIBUTION

This manuscript has not been, and will not be, submitted elsewhere for publication as long as it is under consideration at the *Business Strategy and the Environment*. All authors have contributed significantly to the work, have seen and approved the contents of the manuscript, and agree with its submission to the *Business Strategy and the Environment*.

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